[8] Requirements

Project / **MAIN OBJECTIVE**	Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering.	**MEET**	The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector.

1	The project shall contain very small integrated circuits. Pitch less than 1mm	**MEET**	IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95 IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27
2	The project shall contain 0805 SMT passives components.	**MEET**	Device contain multiple 0805 SMT elements.
3	The project shall contain multi-pin integrated circuits to verify soldering precision.	**MEET**	IC: Shift register : STP16CP05MTR : SO-24 : 24 pins IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins
4	The project shall contain an UPDI programming interface for firmware upload and debugging.	**MEET**	uC Attiny204 with UPDI programming interface
5	Possibility to program using Microchip enviroment and pickit-4 programmer.	**MEET**	It is possible to program the device using pickit-4 and MPLAB software.
6	Shall contain at least two status led.	**MEET**	Status LED'S : LED1, LED2.
7	Shall be able to interact with user.	**MEET**	The user can change the light effect using a capacitive touch sensor.
8	Shall be easy to power up.	**MEET**	Powered by USB-C connector with general mode
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Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 5ba43f3
	Board Name: **3ART12 TOUCH**		Project Name: **Sienkiewicz**	
Sheet Title: Requirements	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1+ (Unreleased)
Sheet Path: /Requirements/		Reviewer:	Size: A4	Sheet: 8 of 9